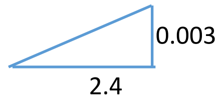
2	**(a)(i)**	Single slit: sin $\theta = \lambda/b$ using small angle approximation sin $\theta \approx$ tan θ  $sin\theta = \frac{0.003}{2.4} = \frac{\lambda}{b}$ (Accept 0.0325) $b = 5.04 \times 10^{-4}\ m$ (accept 4.65×10^{-4}m)	**C1** **A1**
	(a)(ii)1.	(When B decreases↓ , then sinθ increases ↑, and hence θ increases↑, therefore) this results in increase in width of central maximum and the maxima are further apart. The central maximum is also lower in intensity.	**B1** **B1**
	(a)(ii)2.	(When wavelength λ decreases↓, sinθ decreases↓, and hence θ decreases↓) Therefore there is reduced width of central maximum and the maxima are closer. Intensity of the central maximum is unchanged (bright and dark regions)	**B1** **B1**
	(b)(i)	$\theta = \frac{1}{2}(188 - 160) = 14°$ $d \sin\theta = n\lambda \rightarrow d \sin(14°) = 2(633 \times 10^{-9})$ $d = 5.23 \times 10^{-6}$ m $d = \frac{1}{N} \rightarrow N = \frac{1}{d} = 1.91 \times 10^{5}$	**C1** **M1** **C1** **A1**

3	(a)	Electric potential at a point is the work done per unit positive charge in bringing	B1